

Newington College HSC Mini Examination April 2012

Answer Questions 1 to 10 (multiple choice) on the sheet attached to the end of the paper which is to be removed.

Question 1

The correct equation for a circle with centre $(-1, 2)$ and radius of 5 is:

- A. $(x-1)^2 + (y+2)^2 = 25$ B. $(x-1)^2 + (y+2)^2 = 5$
C. $(x+1)^2 + (y-2)^2 = 25$ D. $(x+1)^2 + (y-2)^2 = 5$

Question 2

The integral of $\int \frac{1}{x^2} + \frac{1}{x} dx$ is

- A. $-\frac{3}{x^3} - \frac{2}{x^2} + c$ B. $-\frac{1}{x} + \log_e x + c$
C. $-\frac{1}{x} - \frac{2}{x^2} + c$ D. $-\frac{1}{3x^3} + \log_e x + c$

Question 3

If $\log_a 2 = x$ and $\log_a 3 = y$, then $\log_a 12$ can be written as

- A. $2x + y$ B. $x^2 y$
C. $2xy$ D. $x + 2y$

Question 4

For the parabola, $(y-k)^2 = 4a(x-h)$, the axis of symmetry is given by:

- A. $x = h$ B. $x = k$
C. $y = h$ D. $y = k$

Question 5

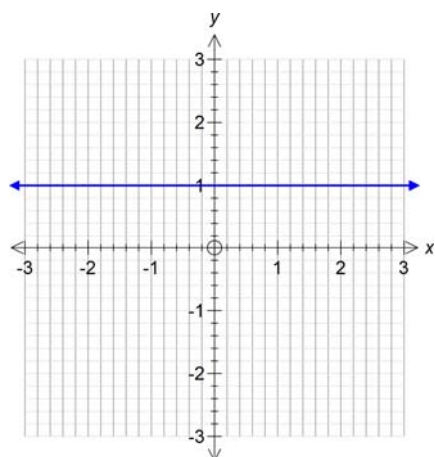
If $f'(x) = -\frac{1}{x^2}$ and $f''(x) = \frac{2}{x^3}$, then for $x > 0$, $f(x)$ is

- A. Increasing and concave down B. Decreasing and concave down
C. Increasing and concave up D. Decreasing and concave up

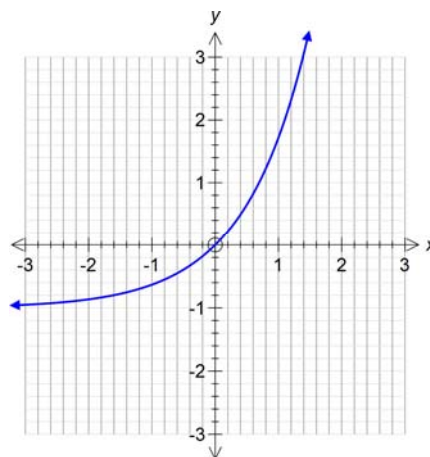
Question 6

Which graph best represents the equation, $y = e^x - e^{-x}$?

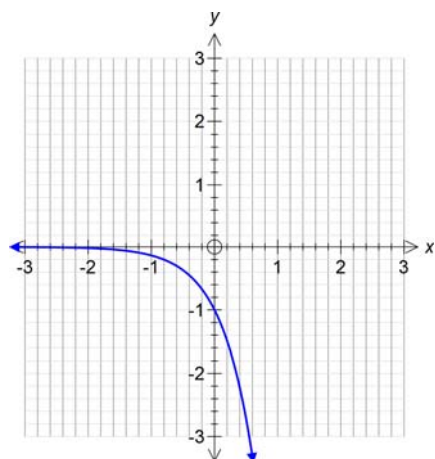
A.



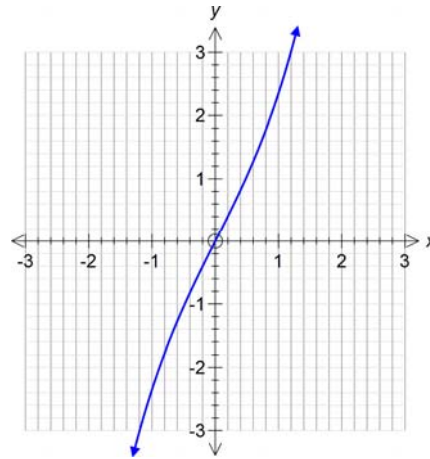
B.



C.



D.



Question 7

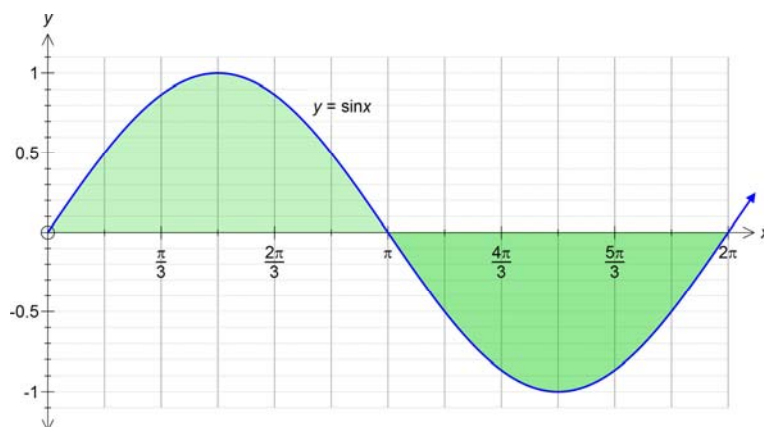
If $\sin \theta = \cos\left(\frac{3\pi}{4}\right)$, then θ could be equal to

- A. $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ B. $\frac{3\pi}{4}$ and $\frac{5\pi}{4}$
- C. $\frac{5\pi}{4}$ and $\frac{7\pi}{4}$ D. $\frac{\pi}{4}$ and $\frac{7\pi}{4}$

Question 8

$2^{-\log_2 5}$ is equal to

- A. $\frac{1}{5}$ B. $-\frac{1}{5}$ C. 5 D. -5

Question 9

Using the graph above, or otherwise, if $\int_0^{\pi} \sin x \, dx = 1$ then the area shaded above is equal to:

- A. 2 B. 0 C. π D. 2π

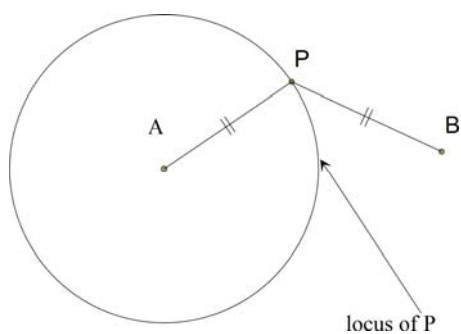
Q10...cont/page 4

Question 10

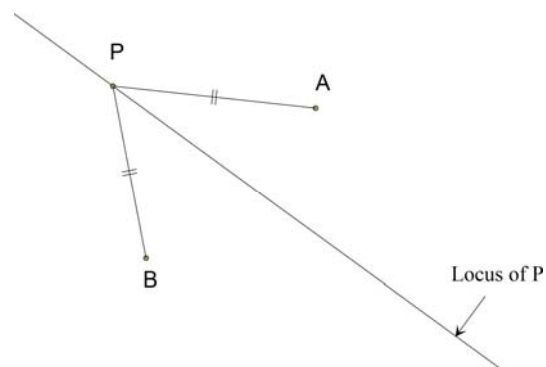
Choose the locus from the diagrams below that is best described by the information given;

“A point P moves such that it is equidistant from two fixed points A and B.”

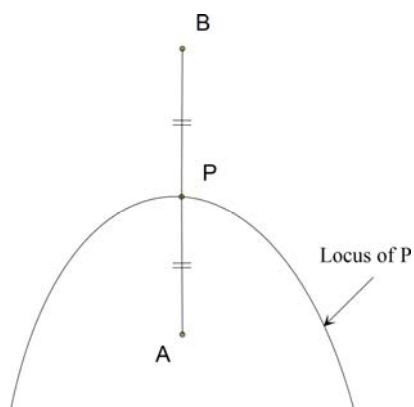
A.



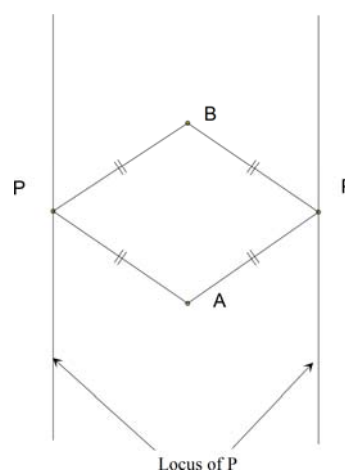
B.



C.



D.



Q11...cont/page 5

Question 11 Start this question in a new booklet (15 Marks) Marks

(a) If α and β are the roots of the quadratic equation, $y = x^2 - 5x + 6$, find:

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\alpha^2 + \beta^2$

(iv) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ (6)

(b) Show that $y = -(x^2 - 3x + 6)$ is negative definite for all values of x . (2)

(c) For the curve, $y = x^2 - 4x$, rewrite in the form, $(x - h)^2 = 4a(y - k)$, and hence, or otherwise, find:

(i) the vertex

(ii) the focal length

(iii) the focus

(iv) the equation of the directrix (7)

Question 12 Start this question in a new booklet (15 marks)

(a) Find the following limits:

(i) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

(ii) $\lim_{x \rightarrow \infty} \frac{1 - x}{x}$ (3)

Question 12 (cont.)**Marks**

(b) Differentiate the following:

(i) $(x^3 - 5)^7$

(ii) $x^3(2 - x)^4$

(iii) $\frac{x}{x^2 - 4}$ (8)

(c) (i) Find the roots of the curve, $y = x^2 - 3x + 2$.

(ii) Find the equation of the tangent for the root with the greatest value of x . (4)

Question 13 Start this question in a new booklet (15 marks)

(a) For the curve, $y = x^3 + x^2 - x + 5$,

(i) find any stationary points and their nature.

(ii) find any points of inflexion.

(iii) Sketch the curve showing all the above and the y intercept. (8)

(b) A square-based prism has a total surface area of 96 cm^2 .

(i) Using $x \text{ cm}$ as the base length and $y \text{ cm}$ as the height draw a diagram of the prism.

(ii) Show that $y = \frac{(48 - x^2)}{2x}$

(iii) Hence, write an equation for the volume of the prism, in terms of x **only**.

(iv) Hence, or otherwise, find the maximum volume of the prism and the values of x and y when this occurs. (7)

Question 14 Start this question in a new booklet**(15 marks)****Marks**

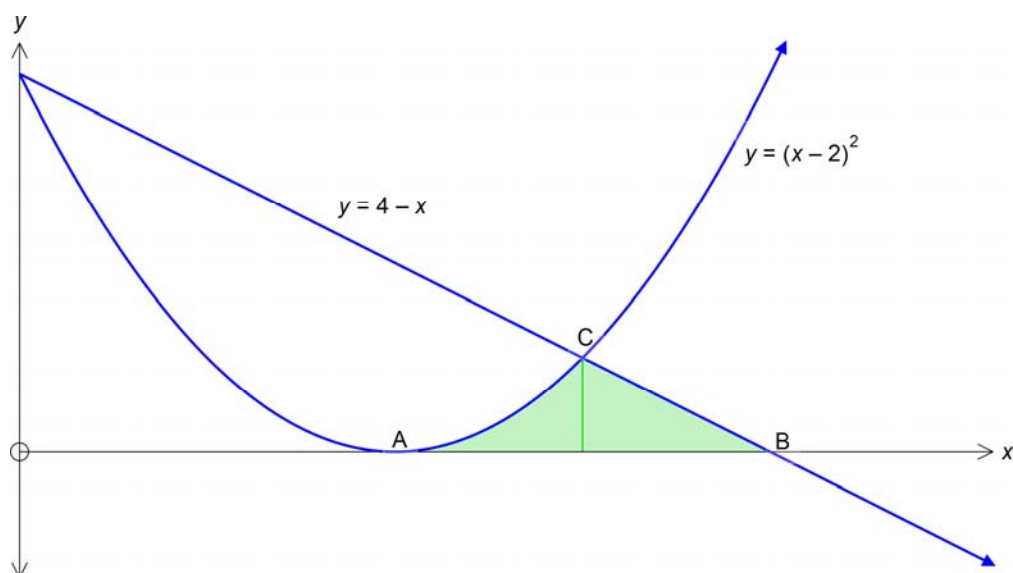
- (a) Find the integral of

(i) $\int 3x^2 + \frac{2}{x^2} dx$

(ii) $\int_0^1 (4-x)^5 dx$ (4)

- (b) Find the area bounded by the curve $y = 4x - x^2$ and the x axis. (3)

- (c) Below are the curve $y = (x-2)^2$ and the straight line $y = 4 - x$,



- (i) Find the points of intersection A, B and C.
(ii) Hence, find the shaded area. (6)

- (d) Without the use of a sketch, explain with reasoning, why $\int_{-a}^a x^5 - x^3 dx = 0$ (2)

Question 15 Start this question in a new booklet (15 marks) Marks

(a) Simplify $\log_3 27 - \log_9 \left(\frac{1}{3}\right) + 7$ (2)

(b) (i) Find the first and second derivatives of $f(x) = \frac{x}{e^x}$.

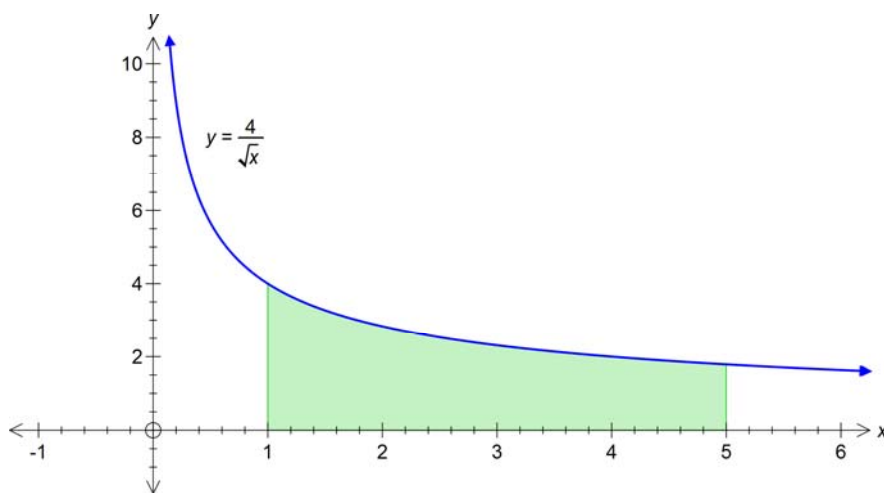
(ii) Find any stationary points for the curve determine their nature.

(iii) Find any points of inflexion.

(iv) Explain why $f(x) \rightarrow 0$ as $x \rightarrow \infty$.

(v) Sketch $f(x) = \frac{x}{e^x}$, using the information above. (8)

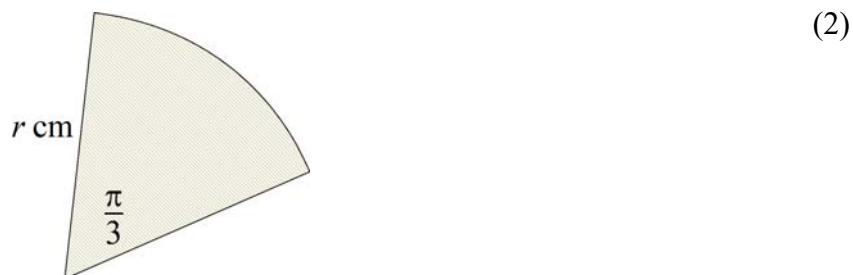
(c) The graph shows the curve $y = \frac{4}{\sqrt{x}}$, with the area under the curve from $1 \leq x \leq 5$ shaded.



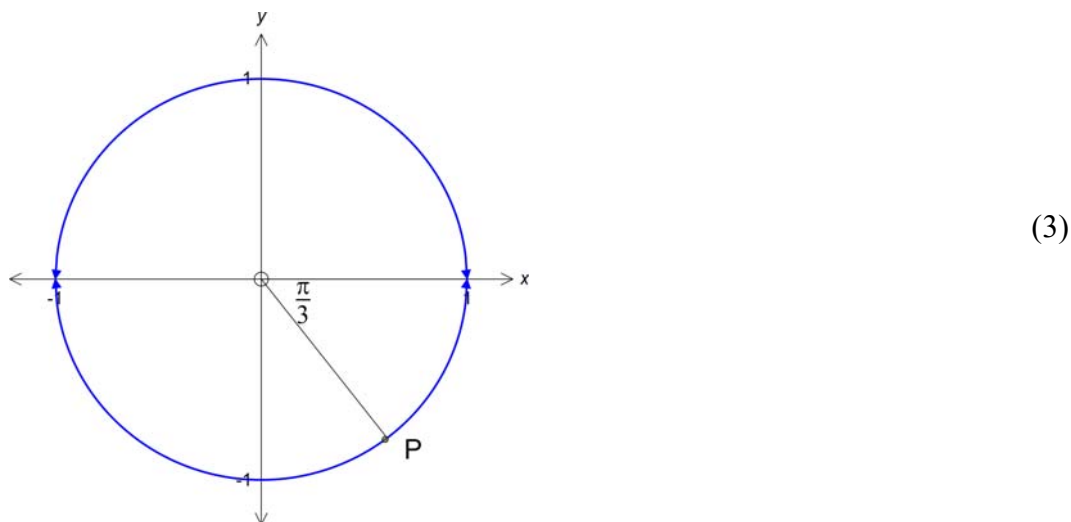
If this area is now revolved around the x axis find the exact value of the volume generated. (5)

Question 16 Start this question in a new booklet (15 marks) Marks

- (a) In the diagram, the area of the sector is $\frac{3\pi}{2} \text{ cm}^2$. Find the radius of the sector.



- (b) In the unit circle shown, find the exact value of the co-ordinates of the point P.



- (c) Solve the following equation, for $0 \leq x \leq 2\pi$,

$$2 \sin^2 x - 1 = 0$$

- (d) (i) Sketch the graph, $y = 2 \sin \frac{x}{2}$, from $0 \leq x \leq 2\pi$
- (ii) Use Simpson's Rule and 5 function values to find the approximate area bounded by the curve and the x-axis.

END OF EXAMINATION

Question 1

C

Question 2

B.

Question 3

A

Question 4

D

Question 5

D

Question 6

D

Question 7

C

Question 8

A

Question 9

A

Question 10

B

Question 11 **Start this question in a new booklet** **(15 Marks)** **Marks**

(a) (i) $\alpha + \beta = -\frac{-5}{1} = 5$ [1]

(ii) $\alpha\beta = \frac{6}{1} = 6$ [1]

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $\quad \quad = 5^2 - 2 \times 6$
 $\quad \quad = 13$ [2]

(iv) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$ [2]
 $\quad \quad = \frac{13}{36}$

(b) $y = -(x^2 - 3x + 6)$ is **negative definite** if for the general equation:

$$a < 0 \text{ and } \Delta < 0$$

In this equation $a = -1 < 0$ so true, and, $\Delta = 3^2 - 4(-1)(-6) = 9 - 24 = -13 < 0$ which is also true so this equation is negative definite. [2]

(c) $y = x^2 - 4x$
 $y + 4 = x^2 - 4x + 4$ [3]
 $y + 4 = (x - 2)^2$

$$(x - 2)^2 = 4\left(\frac{1}{4}\right)(y - (-4))$$

(i) the vertex at $(2, -4)$ [1]

(ii) the focal length = $\frac{1}{4}$ [1]

(iii) the focus at $\left(2, -3\frac{3}{4}\right)$ [1]

(iv) the equation of the directrix is given by $y = -4\frac{1}{4}$ [1]

Question 12 Start this question in a new booklet (15 marks)

(a) (i)
$$\begin{aligned}\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} \\ &= \lim_{x \rightarrow 3} x + 3 \\ &= 6\end{aligned}$$
 [1]

(ii)
$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{1 - x}{x} &= \lim_{x \rightarrow \infty} \frac{1}{x} - 1 \\ &= 0 - 1 \\ &= -1\end{aligned}$$
 [2]

(b) (i)
$$\begin{aligned}\frac{d(x^3 - 5)^7}{dx} &= 3x^2(7)(x^3 - 5)^6 \\ &= 21x^2(x^3 - 5)^6\end{aligned}$$
 [2]

(ii)
$$\begin{aligned}\frac{d[x^3(2 - x)^4]}{dx} &= 3x^2(2 - x)^4 + x^3(-4)(2 - x)^3 \\ &= x^2(2 - x)^3[3(2 - x) - 4x] \\ &= x^2(2 - x)^3(6 - 7x)\end{aligned}$$
 [3]

(iii)
$$\frac{d\left(\frac{x}{x^2 - 4}\right)}{dx} = \frac{(x^2 - 4) - 2x(x)}{(x^2 - 4)^2} = -\frac{x^2 + 4}{(x^2 - 4)^2}$$
 [3]

(c) (i)
$$\begin{aligned}y &= x^2 - 3x + 2 \\ x^2 - 3x + 2 &= 0 \\ (x - 2)(x - 1) &= 0 \\ x &= 1 \text{ or } 2\end{aligned}$$
 [2]

(ii) If $x = 2$, then
$$\begin{aligned}\frac{dy}{dx} &= 2x - 3 \\ \text{If } x &= 2 \text{ then} \\ \frac{dy}{dx} &= 2(2) - 3 = 1 \\ \text{At } x &= 2, y = 2^2 - 3(2) + 2 = 0\end{aligned}$$
 [2]

$$y - 0 = 1(x - 2)$$

$$y = x - 2$$

Question 13 Start this question in a new booklet (15 marks)

(a) (i) $y = x^3 + x^2 - x + 5$ If $\frac{dy}{dx} = 0$, then $x = \frac{1}{3}$ or -1

$$\frac{dy}{dx} = 3x^2 + 2x - 1$$

$$= (3x - 1)(x + 1)$$

Hence, stationary points at $(\frac{1}{3}, 4\frac{22}{27})$ and $(-1, 6)$

Since, $y = \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2 - \frac{1}{3} + 5 = 4\frac{22}{27}$, and

$$y = (-1)^3 + (-1)^2 - (-1) + 5 = 6$$

x	-2	-1	0	$\frac{1}{3}$	1
$\frac{dy}{dx}$	+ve	0	-ve	0	+ve
slope	/	—	\	—	/
		Max tp		Min tp	

[3]

(ii) $y = x^3 + x^2 - x + 5$ If $\frac{d^2y}{dx^2} = 0$, then $x = -\frac{1}{3}$

$$\frac{dy}{dx} = 3x^2 + 2x - 1$$

$$\frac{d^2y}{dx^2} = 6x + 2$$

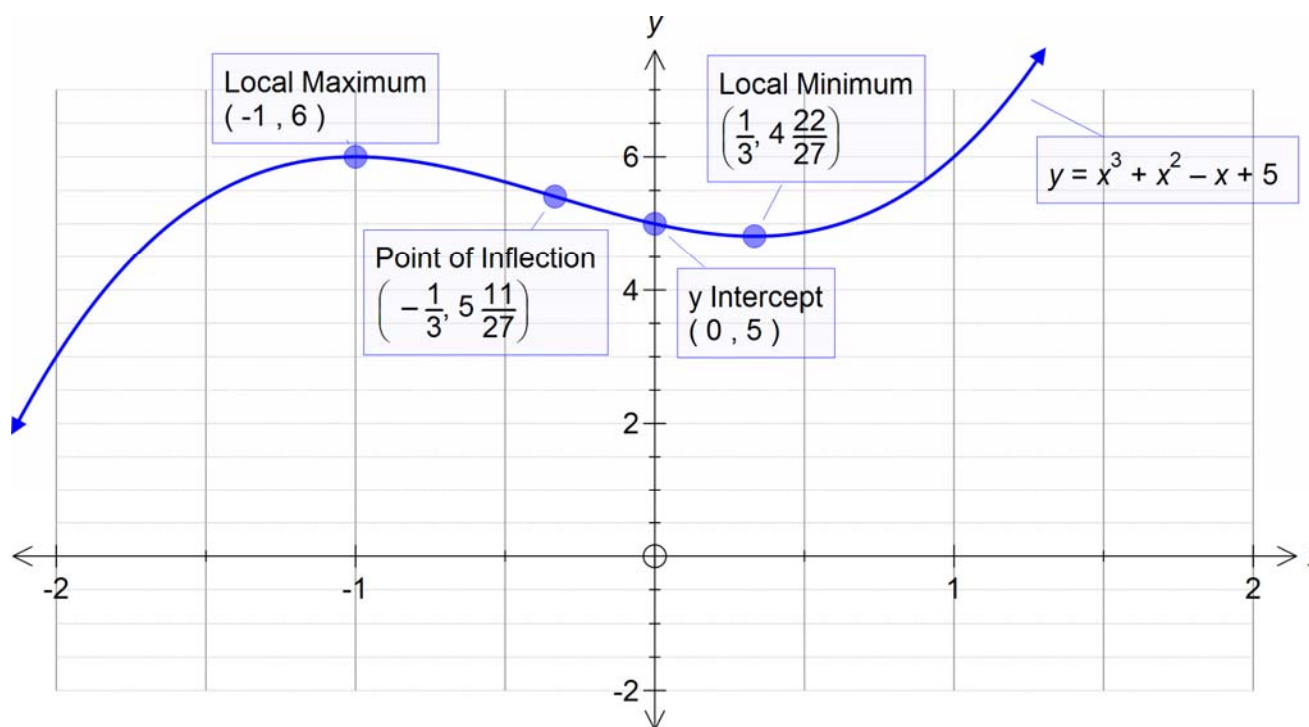
$$y = \left(-\frac{1}{3}\right)^3 + \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) + 5 = 5\frac{11}{27}$$

x	-1	$-\frac{1}{3}$	0
$\frac{dy}{dx}$	-ve	0	+ve
concavity	down		up
		Pt of I	

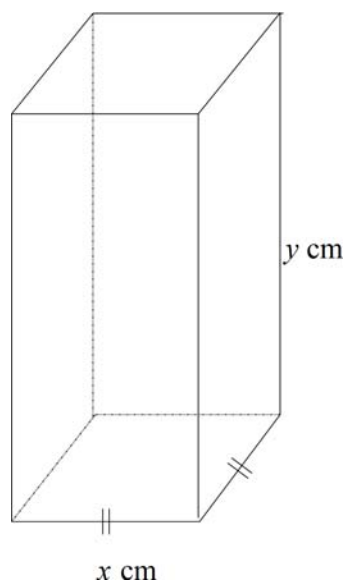
Hence, point of inflexion at $(-\frac{1}{3}, 5\frac{11}{27})$

[2]

(iii)



(b) (i)



[1]

(ii) Surface Area = 96 cm^2 . Hence,

$$2x^2 + 4xy = 96$$

$$4xy = 96 - 2x^2$$

$$y = \frac{2(48 - x^2)}{4x}$$

$$y = \frac{48 - x^2}{2x}$$

[2]

(iii)

$$\text{Volume} = x^2 y$$

$$V = \frac{x^2(48 - x^2)}{2x} = \frac{1}{2}x(48 - x^2) \quad [1]$$

$$V = 24x - \frac{x^3}{2}$$

(iv) Now, maximum volume when $\frac{dV}{dx} = 0$,

$$V = 24x - \frac{x^3}{2} \quad [3]$$

$$\frac{dV}{dx} = 24 - \frac{3x^2}{2}$$

$$3x^2 = 48$$

$$x^2 = 16$$

$$x = 4 \text{ cm}, \quad x > 0$$

$$y = \frac{48 - 16}{8} = 4 \text{ cm}$$

$$\text{Max Volume} = 64 \text{ cm}^3$$

Question 14

(a) (i) $\int 3x^2 + \frac{2}{x^2} dx = \int 3x^2 + 2x^{-2} dx$

$$= \frac{3x^3}{3} + \frac{2x^{-1}}{(-1)} + c$$

$$= x^3 - \frac{2}{x} + c$$

[1]

(ii) $\int_0^1 (4-x)^5 dx = \left[\frac{(4-x)^6}{(-1)(6)} \right]_0^1$

$$= -\frac{1}{6} \left[(4-x)^6 \right]_0^1$$

$$= -\frac{1}{6} (3^6 - 4^6)$$

$$= \frac{3367}{6}$$

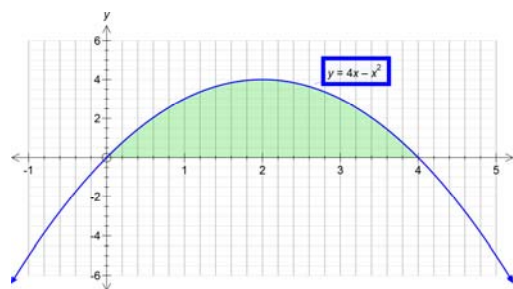
[2]

- (b) Find the area bounded by the curve $y = 4x - x^2$ and the x axis.

Area wholly above the x-axis, hence

$$\begin{aligned} \text{Area} &= \int_0^4 4x - x^2 \, dx \\ &= \left[2x^2 - \frac{x^3}{3} \right]_0^4 \\ &= \left(32 - \frac{64}{3} \right) \\ &= \frac{32}{3} \text{ units}^2 \end{aligned}$$

[3]



- (c) (i) At A, for the curve, $y = (x-2)^2$, $y = 0$.

So, $x = 2$, i.e. A (2, 0)

At B, for the line, $y = 4 - x$, $y = 0$

So, $x = 4$, i.e. B (4, 0)

At C, the curves $y = (x-2)^2$ and $y = 4 - x$ intersect, so,

$$(x-2)^2 = 4 - x$$

$$x^2 - 4x + 4 = 4 - x$$

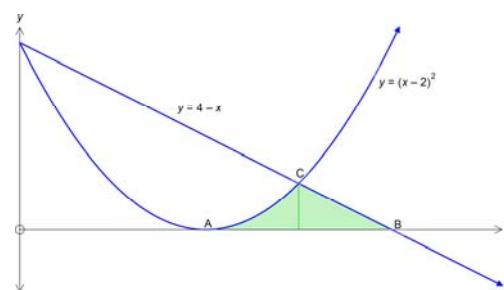
$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$x = 0 \text{ or } 3, x > 0$$

$$x = 3$$

If $x = 3$ then $y = 1$, so C (3, 1) [3]



- (ii)

$$\begin{aligned} \text{Area} &= \int_2^3 (x-2)^2 \, dx + \int_3^4 4 - x \, dx \\ &= \left[\frac{(x-2)^3}{3} \right]_2^3 + \left[4x - \frac{x^2}{2} \right]_3^4 \\ &= \left(\frac{1}{3} - 0 \right) + \left[(16 - 8) - \left(12 - \frac{9}{2} \right) \right] \\ &= \frac{1}{3} + 8 - \frac{15}{2} \\ &= \frac{5}{6} \text{ square units} \end{aligned} \quad [3]$$

(d) $\int_{-a}^a x^5 - x^3 \, dx = 0$. The graph of $y = x^5 - x^3$ is an odd function, i.e.

$$f(a) = -f(-a), \text{ and as such has point symmetry about the origin.}$$

Thus the areas above and below the x -axis on either sides of the y -axis must be equal, hence the integral:

$$\int_{-a}^a x^5 - x^3 \, dx = 0 \quad [2]$$

Question 15**(15 marks)****Marks**

(a) $\log_3 27 - \log_9 \left(\frac{1}{3} \right) + 7 = \log_3 (3)^3 - \log_9 \left(9^{-\frac{1}{2}} \right) + 7$

$$= 3 \log_3 3 + \frac{1}{2} \log_9 9 + 7$$

$$= 10 \frac{1}{2} \quad [2]$$

(b) (i) $f(x) = \frac{x}{e^x}$

$$f'(x) = \frac{e^x(1) - x(e^x)}{e^{2x}}$$

$$= \frac{e^x(1-x)}{e^{2x}} \quad [2]$$

$$= \frac{1-x}{e^x}$$

$$f''(x) = \frac{e^x(-1) - (1-x)(e^x)}{e^{2x}}$$

$$= \frac{e^x(x-2)}{e^{2x}}$$

$$= \frac{x-2}{e^x}$$

(ii) If $f'(x) = 0$
 $1 - x = 0$
 $x = 1$

If $x = 1$ then $y = e^{-x}$ and $f''(x) = \frac{-1}{e^x} < 0$. [2]

Hence, $(1, e^{-x})$ is a max turning point.

(iii) If $f''(x) = 0$
 $x - 2 = 0$
 $x = 2$

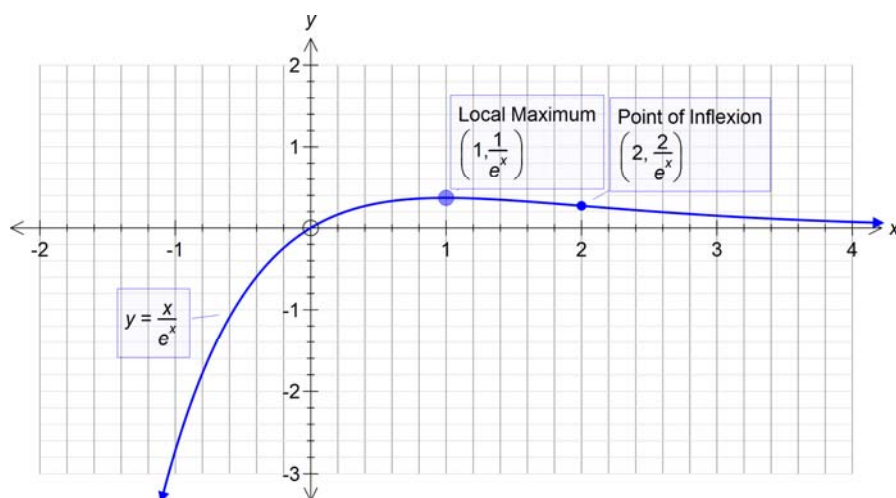
If $x = 2$ and $y = 2e^{-x}$

x	1	2	3
$f''(x)$	-ve	0	+ve
	Concave down		Concave up

Hence point of inflexion at $(2, 2e^{-x})$

(iv) Let $y = xe^{-x}$, so $e^{-x} \rightarrow 0$, as $x \rightarrow \infty$ [1]

(v)



[3]

(c)

$$y = \frac{4}{\sqrt{x}}$$

$$y^2 = \frac{16}{x}$$

$$\text{Volume} = \pi \int_1^5 \frac{16}{x} dx$$

$$= \pi [16 \log_e x]_1^5$$

$$= 16\pi (\log_e 5 - \log_e 1)$$

$$= 16\pi \log_e 5$$

[5]

Question 16

(a)

$$\text{Area} = \frac{r^2 \theta}{2}$$

$$\frac{3\pi}{2} = \frac{r^2 \frac{\pi}{3}}{2}$$

$$r^2 = 9$$

$$r = 3$$

[2]

(b)

$$\text{At } P, \quad \left(\cos\left(-\frac{\pi}{3}\right), \sin\left(-\frac{\pi}{3}\right) \right) = \left(\cos\frac{\pi}{3}, -\sin\frac{\pi}{3} \right)$$

$$= \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right)$$

[3]

(c)

$$2 \sin^2 x - 1 = 0$$

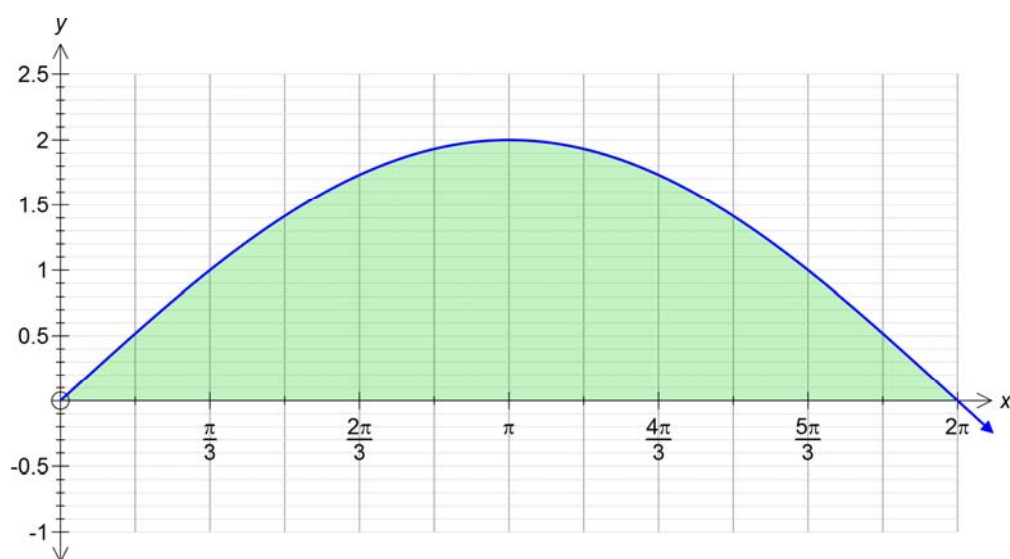
$$\sin^2 x = \frac{1}{2}$$

$$\sin x = \frac{1}{\sqrt{2}} \quad \text{or} \quad -\frac{1}{\sqrt{2}}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4} \quad \text{or} \quad \frac{7\pi}{4}$$

[3]

(d) (i)



[3]

(ii)
$$h = \frac{2\pi - 0}{4} = \frac{\pi}{2}$$

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x) = 2\sin\frac{x}{2}$	0	$\sqrt{2}$	2	$\sqrt{2}$	0
Factor	x1	x4	x2	x4	x1
Value	0	$4\sqrt{2}$	4	$4\sqrt{2}$	0

$$\text{Area} = \frac{\pi}{6} (4 + 8\sqrt{2}) \approx 8$$

[5]